products further would force us to integrate multiple data sources, even for the most straightforward use cases.

Imagine you create two data products: Subscriber Details and Home Address. In the Home Address data product, you store country, state, street, and home number. However, you cannot use the Home Address data product on its own. It has no value without knowledge about subscriber details. In this case, we should have a single data product, keeping both datasets.

This heuristic implies that the smallest viable data product is the size of the domain dataset. This could often be a good default solution, but you need to consider its feasibility. For example, managing thousands of data products might be more complex and less effective than doing the same for hundreds of them.

Avoid data products with a canonical model of the whole enterprise

The canonical model of the entire enterprise usually leads to losing the context of the stored information and architecture of the centralized data warehouse responsible for storing that canonical model. This design is the reverse of a data mesh architecture that is meant to be decentralized.

Align with usage contract

Datasets in a data product should be shared under the same usage contract. Because a data product will be an autonomous component in most cases, it will be much easier to enforce contracts and authorization on the data product level.

An example of such a contract is personal data under the European's Union GDP. We should separate the Subscriber data product (which stores such data). Otherwise, any less regulated data combined would be hidden behind stringent security and privacy rules.

Align with organizational boundaries